Type MS · Midpoints and points that split a line in a ratio

Answer 1 of 10 · Question 6 part a · 2 marks

2024 JUN · Paper 01 · extract 1 of 5 · find part a in this mark scheme extract

Question	Scheme	Marks
6 (a)	$p=1,\ \ q=8$	B1B1 [2]
(b)	$\left(\text{Gradient of } l=\dfrac{12-2}{3--2}=\right)2$	B1
	(Gradient of the perpendicular =) $-\dfrac{1}{"2"}$	B1ft
	Equation of k	
	$y-"8"="-\dfrac{1}{2}"(x-"1")$	M1
	$\Rightarrow 2y+x-17=0$ *	A1 cso [4]
(c)	When $y=0$, $(2\times 0+x-17=0\Rightarrow)\ x=17$	B1
	$(\lvert CD\rvert=)\sqrt{(0-"8")^2+("17"-"1")^2}=\sqrt{320}=\left[8\sqrt{5}\right]$	M1A1 [3]
(d)	(Length of CX =) $\dfrac{2\times 80}{8\sqrt{5}}\left(=4\sqrt{5}\right)$	B1
	$("4\sqrt{5}")^2=(m-"1")^2+(n-"8")^2$	M1
	$"2"=\dfrac{n-"8"}{m-"1"}\Rightarrow[n=2m+6]$ oe **or** $y-2="2"(x+2)\Rightarrow[y=2x+6]$	M1
	or $y-12="2"(x-3)\Rightarrow[y=2x+6]$ oe	
	$\Rightarrow "80"=(m-"1")^2+(2m+6-"8")^2\ (\Rightarrow 0=5m^2-10m-75$ oe eg $0=m^2-2m-15)$	ddM1
	$\Rightarrow(5m+15)(m-5)=0$ oe eg $(m+3)(m-5)=0$	M1
	$m=5,(-3)$	A1
	$n(=2m+6=2\times 5+6)=16$	A1 [7]
ALT1	(Length of CX =) $\dfrac{2\times 80}{8\sqrt{5}}\left(=4\sqrt{5}\right)$	B1
	$(m-"17")^2+(n)^2=("4\sqrt{5}")^2+("8\sqrt{5}")^2$	M1
	$"2"=\dfrac{n-"8"}{m-"1"}\Rightarrow[n=2m+6]$ oe **or** $y-2="2"(x+2)\Rightarrow[y=2x+6]$	M1
	or $y-12="2"(x-3)\Rightarrow[y=2x+6]$ oe	
	$\Rightarrow "400"=(m-"17")^2+(2m+6)^2\ (\Rightarrow 0=5m^2-10m-75$ oe eg $0=m^2-2m-15)$	ddM1
	$\Rightarrow(5m+15)(m-5)=0$ oe eg $(m+3)(m-5)=0$	M1
	$m=5,(-3)$	A1
	$n(=2m+6=2\times 5+6)=16$	A1 [7]

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ALT2	(Length of CX =) $\frac{2\times 80}{8\sqrt{5}}\left(=4\sqrt{5}\right)$	B1
	Length of AB = $\sqrt{(12-2)^2+(3--2)^2}\left(=5\sqrt{5}\right)$	M1
	Vector $\overrightarrow{AB}=\begin{bmatrix}3--2\\12-2\end{bmatrix}\left(=5\begin{bmatrix}1\\2\end{bmatrix}\right)$	M1
	$\left(\left\|\overrightarrow{CX}\right\|=4\sqrt{5}\Rightarrow\right)\overrightarrow{CX}=4\begin{bmatrix}1\\2\end{bmatrix}=\begin{bmatrix}4\\8\end{bmatrix}$	ddM1
	$\Rightarrow$ Coordinates of X are $(1,8)+(4,8)=(5,16)$	M1A1
	$(m=5 \quad n=16)$	A1 [7]
ALT3	$\frac{1}{2}\begin{vmatrix}17 & m & 1 & 17\\0 & n & 8 & 0\end{vmatrix}$	B1
	$\frac{1}{2}\begin{vmatrix}"17" & m & "1" & "17"\\0 & n & "8" & 0\end{vmatrix}=80\Rightarrow\frac{1}{2}\left(17n+"8"m-("136"+n)\right)=80\left(\Rightarrow 2n+m=37\right)$	M1
	$"2"=\frac{n-8}{m-1}\Rightarrow[n=2m+6]$ oe **or** $y-2="2"(x+2)\Rightarrow[y=2x+6]$ oe **or** $y-12="2"(x-3)\Rightarrow[y=2x+6]$ oe	M1
	$\frac{1}{2}\begin{vmatrix}"17" & m & "1" & "17"\\0 & "2m+6" & "8" & 0\end{vmatrix}$ or $\frac{1}{2}\left(17("2m+6")+"8"m-("136"+"2m+6")\right)=80$	ddM1
	or $2("2m+6")+m=37\Rightarrow m=$	M1
	$m=5,\ n=16$	A1A1 [7]
		Total 16 marks

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Part	Mark	Notes
(a)	B1	For $p = 1$ OR $q = 8$ Allow anything that clearly implies these values eg (1, 8) or 1, 8
	B1	For $p = 1$ AND $q = 8$ Allow anything that clearly implies these values eg (1, 8) or 1, 8
(b)	B1	For the gradient of $l = 2$
	B1ft	For the gradient of $k = -\frac{1}{"2"}$. Ft their calculation of the gradient of l.
	M1	For the correct unsimplified equation for k, using their p and q and any **changed** gradient. If the candidate uses $y = mx + c$, a fully correct rearrangement to find c must be shown (this can be implied by a correct c for their equation) and concluded with the equation of the line written.
	A1 cso	For the equation of k in the required form. Minimum steps shown, no errors or omissions. This must from a correct p and q
Watch for methods using vectors. If the candidate arrives at the correct unsimplified equation with no obviously incorrect work, following vector work, allow B1 B1ft M1 and the final A1 if the equation is given in the correct form. (See example in Practice)		
(c)	B1	For the x coordinate of $D = 17$
	M1	For a correct **method** to find the length of CD using their p and q and their 17
	A1	For the correct length of $\sqrt{320}$ oe
(d)	B1	Any correct unsimplified calculation for CX
	M1	For correctly using Pythagoras to form an equation in terms of m and n or x and y. Follow through their CX (ie coming from correct calculation with their p, q, their point D).
	M1	For an unsimplified equation in terms of m and n or x and y correctly using the gradient of l. Follow through their gradient for l, their p and their q. Allow working in p, q or x, y Note this work is sometimes being seen in (b). **As long as it is used in part (d),** the equation found in (b) may be given credit.
	ddM1	For eliminating either m or n from either equation and forming a 3TQ. Allow errors in processing if an initial correct method following from 2 correct unsimplified equations to eliminate one of the variables is seen. Allow working in p, q or x, y Dependent on both previous method marks. Note, the previous M mark is for the unsimplified equation, so if simplified incorrectly, this can be used here to gain this mark (general principle of marking).
	M1	For a minimally acceptable attempt to solve (see general guidance) their 3TQ This mark can be implied from a correct value of m = 5. If the quadratic is not the correct quadratic, method must be shown.
	A1	For one of $m = 5$ or $n = 16$
	A1	For both $n = 16$ and $m = 5$
ALT1	First B, final 2 A marks – as main scheme	
	M1	For correctly using Pythagoras to form an equation in terms of m and n or x and y. Follow through their CX and CD (ie coming from correct calculation with their p, q, their point D).
	M1	For an unsimplified equation in terms of m and n or x and y correctly using the gradient of l. Follow through their gradient for l, their p and their q. Allow working in p, q or x, y Note this work is sometimes being seen in (b). **As long as it is used in part (d),** the equation found in (b) may be given credit.
	ddM1	For eliminating either m or n from either equation and forming a 3TQ. Allow errors in processing if an initial correct method following from 2 correct unsimplified equations to eliminate one of the variables is seen. Allow working in p, q or x, y Dependent on both previous method marks. Note, the previous M mark is for the unsimplified equation, so if simplified incorrectly, this can be used here to gain this mark (general principle of marking).
	M1	For a minimally acceptable attempt to solve (see general guidance) their 3TQ This mark can be implied from a correct value of m = 5. If the quadratic is not the correct quadratic, method must be shown.

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ALT2	B1	Any correct unsimplified calculation for CX
	M1	Correct method to find the length of AB
	M1	Correct method to find vector AB
	ddM1	Correctly deduces the relationship between their vectors CX and AB Dependent on both previous method marks.
	M1	Correct method to find the coordinates of X using their vector work.
	A1	For one of $m = 5$ or $n = 16$
	A1	For both $n = 16$ and $m = 5$
ALT3	B1	For a fully correct array as shown oe
	M1	For placing their array (though ft their p and q and their "17") = 80 and correctly multiplying out the discriminant to form an unsimplified equation in m and n
	M1	For an unsimplified equation in terms of m and n or x and y correctly using the gradient of l. Follow through their gradient for l, their p and their q. Allow working in p, q or x, y Note this work is sometimes being seen in (a). **As long as it is used in part (d),** the equation found in (a) may be given credit.
	ddM1	For eliminating either m or n seen either in the array or in their equation formed. Allow errors in processing if an initial correct method following from 2 correct unsimplified equations to eliminate one of the variables is seen. Allow working in p, q or x, y Dependent on both previous method marks. Note, the previous M mark is for the unsimplified equation, so if simplified incorrectly, this can be used here to gain this mark (general principle of marking).
	M1	For solving their linear equation, allow one error in processing, leading to a value of m or n
	A1	For one of $m = 5$ or $n = 16$
	A1	For both $n = 16$ and $m = 5$
If a partial method is seen, with correct answers not from obvious incorrect working, marks may be given (general point of marking unless precluded from the mark scheme for a question). If a partial method is seen and marks cannot be fitted to the main or ALT schemes – send to review.		

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Answer 1 of 10 · Question 6 part a · 2 marks

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Question	Scheme	Marks

Type MS · Midpoints and points that split a line in a ratio

Answer 2 of 10 · Question 11 part b · 2 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 4 · find part b in this mark scheme extract

Number		
11 (a)	$\frac{y-6}{x-5}=\frac{6-3}{5--1}\Rightarrow 2y-12=x-5$	M1A1 [2]
(b)	$\left(\frac{2\times5+1\times-1}{2+1},\frac{2\times6+1\times3}{3}\right)\Rightarrow(3,\ 5)$*	M1A1 [2]
(a)		
M1	A fully correct method for finding the equation of a straight line e.g. $\frac{y-y_1}{x-x_1}=\frac{y_2-y_1}{x_2-x_1}$	
A1	$2y-12=x-5$ oe	
(b)		
M1	Use of $\left(\frac{qx_1+px_2}{p+q},\frac{qy_1+py_2}{p+q}\right)$ or $\begin{pmatrix}-1\\3\end{pmatrix}+\frac{2}{3}\times\begin{pmatrix}6\\3\end{pmatrix}$	
A1	Obtains the **given** coordinates	

Type MS · Midpoints and points that split a line in a ratio

Answer 2 of 10 · Question 11 part b · 2 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 4 · find part b in this mark scheme extract

(c)	Gradient of perpendicular to AB = -2	B1
	$\frac{5-n}{3-m}=-2\Rightarrow(11=2m+n)$	M1
	Radius = 5, so the length of AC = 10	
	$100=(3-n)^2+(-1-m)^2\Rightarrow(90=n^2-6n+m^2+2m)$	M1
	Solves simultaneous equations $90=121-44m+4m^2-6(11-2m)+m^2+2m$	depM1
	$35=5m^2-30m\Rightarrow m^2-6m-7=0$	depM1
	$(m-7)(m+1)=0\Rightarrow m=7,\ n=-3$	A1 [6]
ALT (c) using vectors		
(c)	$\overrightarrow{AB}=\begin{pmatrix}5\\6\end{pmatrix}-\begin{pmatrix}-1\\3\end{pmatrix}=\begin{pmatrix}6\\3\end{pmatrix}\Rightarrow$ Perpendicular vector to $\overrightarrow{AB}=\begin{pmatrix}3\\-6\end{pmatrix}$	B1
	$\lvert\overrightarrow{AB}\rvert=\sqrt{45}=3\sqrt{5}\Rightarrow\lvert AP\rvert=2\sqrt{5}\qquad\lvert\overrightarrow{AC}\rvert=10$	M1M1
	Using Pythagoras $\lvert\overrightarrow{PC}\rvert=\sqrt{100-20}=\sqrt{80}=4\sqrt{5}$	depM1
	$\overrightarrow{PC}=\frac{4\sqrt{5}}{3\sqrt{5}}\times\begin{pmatrix}3\\-6\end{pmatrix}=\begin{pmatrix}4\\-8\end{pmatrix}$	depM1
	$\Rightarrow$ Coordinates of C are $(3+(4),\ 5+(-8))\Rightarrow(7,-3)$	A1 [6]

(c)

B1 Gradient of perpendicular to AB = -2 (Can be implied by M1)

M1 Obtains an equation using the gradient of the perpendicular and the points (m, n) and P (Condone if given in terms of x and y)

M1 Obtains a second equation using AC = 10 and the points A and C

depM1 Solve simultaneously to obtain a 3 term quadratic

depM1 Solve their 3 term quadratic

A1 $m = 7,\ n = -3$

ALT:

B1 Perpendicular vector to $\overrightarrow{AB}=\begin{pmatrix}3\\-6\end{pmatrix}$

M1 Finds $\lvert\overrightarrow{AB}\rvert$ or $\lvert\overrightarrow{AP}\rvert$ or $\overrightarrow{PC}=\begin{pmatrix}3\\5\end{pmatrix}+\lambda\begin{pmatrix}3\\-6\end{pmatrix}$

M1 States $\lvert\overrightarrow{AC}\rvert=10$ or $\lambda=\frac{4}{3}$

depM1 Use of Pythagoras to find $\lvert\overrightarrow{PC}\rvert$ or $\overrightarrow{PC}=\begin{pmatrix}3\\5\end{pmatrix}+\frac{4}{3}\times\begin{pmatrix}3\\-6\end{pmatrix}$

depM1 Finds $\overrightarrow{PC}$

A1 $m = 7,\ n = -3$

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Answer 2 of 10 · Question 11 part b · 2 marks

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(d)	$\frac{y-'-3'}{x-'7'}=\frac{1}{2}\Rightarrow\left\{y=\frac{x-13}{2}\right\},\ \frac{y-3}{x--1}=-2\Rightarrow\{y=-2x+1\}$ Solving simultaneous equations by any method $\frac{x-13}{2}=-2x+1\Rightarrow p=3$ and $q=-5$	M1 M1A1 [3]
ALT (d) using vectors		
(d)	$\overrightarrow{AD}=\overrightarrow{PC}=\begin{pmatrix}4\\-8\end{pmatrix}$ Coordinates of point D $\Rightarrow(-1+(4),\ 3+(-8))=(3,-5)$	M1 M1A1 [3]
(e)	Length of AB $\sqrt{(6-3)^2+(5--1)^2}=3\sqrt{5}$ Length of CD $\sqrt{(-3--5)^2+(7-3)^2}=\sqrt{20}=2\sqrt{5}$ Area of trapezium $=\frac{1}{2}(3\sqrt{5}+2\sqrt{5})\times4\sqrt{5}=50$ (units^2)	M1 A1 M1A1 [4]
ALT (e) using vectors		
	Area $=\frac{1}{2}\begin{vmatrix}-1 & 5 & 7 & 3 & -1\\ 3 & 6 & -3 & -5 & 3\end{vmatrix}$ $=\frac{1}{2}[(-6-15-35+9)-(15+42-9+5)]=50$	 M1A1 M1A1 [4]
		Total 17 marks

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Answer 2 of 10 · Question 11 part b · 2 marks

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Type MS · Midpoints and points that split a line in a ratio

Answer 3 of 10 · Question 9 part b · 2 marks

2024 JUN · Paper 01 (regional R) · find part b in this mark scheme extract

9(a)	$\frac{y-8}{8-3}=\frac{x-6}{6--4}\Rightarrow\frac{y-8}{5}=\frac{x-6}{10}\Rightarrow\left[y-8=\frac{1}{2}(x-6)\right]$	M1A1 [2]
(b)	$C=\left(\frac{4\times6+1\times-4}{4+1},\frac{4\times8+1\times3}{4+1}\right)=(4,7)$	B1,B1 [2]
(c)	$-2=\frac{q-7}{p-4}\Rightarrow q=15-2p$ $8\sqrt{5}=\sqrt{(4-p)^2+(7-q)^2}$ $\Rightarrow 320=(4-p)^2+(2p-8)^2\Rightarrow p^2-8p-48=0$ $\Rightarrow(p+4)(p-12)=0$ $\Rightarrow p=-4\quad q=15-2\times-4=23\Rightarrow D=(-4,23)$	M1 M1 M1 M1 A1A1 [6]
(d)	Area of triangle $ACD=\frac{1}{2}\times(23-3)\times(4--4)=80$ [square units] **ALT 1** $\frac{1}{2}\begin{bmatrix}-4 & 4 & -4 & -4\\ 3 & 7 & 23 & 3\end{bmatrix}=\frac{1}{2}\left[(-28+92-12)-(12-28-92)\right]=80$ **ALT 2** $A=\frac{1}{2}\times AC\times CD=\frac{1}{2}\times4\sqrt{5}\times8\sqrt{5}=80$	M1A1 [2] [M1A1] [M1A1]
		Total 12 marks

Type MS · Midpoints and points that split a line in a ratio

Answer 4 of 10 · Question 8 part c · 2 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

8(a)	$AB^2 = 4^2 + 2^2$, $BC^2 = 2^2 + 6^2$, $AC^2 = 2^2 + 4^2$	M1 (any one)
	(i) $AB = \sqrt{20}$ (4.47) (ii) $BC = \sqrt{40}$ (6.32) (iii) $AC = \sqrt{20}$ (4.47) or equivalents	A1A1A1 (4)
(b)	Any complete method for finding one of the angles: eg $AB^2 + AC^2 = BC^2 \Rightarrow \angle A = 90°$ or use trigonometry	M1
	$\angle A = 90°$, $\angle B = \angle C = 45°$	A1, A1 (3)
I	(centre at midpoint of BC) $(5,5)$	M1A1 (2)
(d)	Radius $= \frac{1}{2}BC = \frac{1}{2}\sqrt{40} = \sqrt{10}$ (Working for (d) may be seen in a previous part)	M1A1 (2) [11]

(a) M1	Use Pythagoras with a plus sign to obtain AC^2, BC^2 or AC^2. If the answer is incorrect it must be clear that the correct coordinates have been used correctly.
A1A1A1	Award A1 for each correct length. Ignore labels (i), (ii) and (iii). Award M1A1A1A1 / M1A1A1A0 / M1A1A0A0 as appropriate.
SC:	If there is no working shown but at least one length is correct, award M1 and deduct one A mark for each incorrect length. (no length correct and no working ⇒ M0) If all 3 lengths are correct to at least 3 sf, award M1A1A1A0 If 2 are correct to at least 3 sf, award M1A1A0A0
(b) M1	Attempt to obtain any of the required angles. Method must be complete (ie reach a value for one angle) and formula used must be correct and values must be substituted into a correct formula.
A1	$\angle A = 90°$ Any labelling given can be ignored.
A1	$\angle B = \angle C = 45°$ All 3 correct w/o working scores M1A1A1
(c) M1	For indicating that the centre is at the midpoint of BC. This can be stated explicitly or used by attempting to find the midpoint. **OR:** Find equations for perpendicular bisectors of 2 of the sides and find the point of intersection
A1	Both coordinates correct. Correct answer written down w/o working scores M1A1
(d) M1	For indicating that the radius is half the length of BC. This can be stated explicitly or used by attempting to find half of their BC (not nec in the required form).
A1	Correct length of the radius, in the required form.
NB	If half the length of BC has been found earlier the marks for (d) can only be awarded if the length of the radius has been written in (d).

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Answer 4 of 10 · Question 8 part c · 2 marks

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Question Number	Scheme	Marks
	$\frac{1}{1}$	

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Answer 5 of 10 · Question 9 part d · 2 marks

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Number		
9 (a)	$a=-1,\ b=-2$	B1,B1 [2]
(b)	Gradient of $l_1=-2,\Rightarrow$ Gradient of $l_2=\frac{1}{2}$	B1,B1
	$180=(x+1)^2+(y-6)^2$	M1
	$\frac{1}{2}=\frac{y-6}{x+1}\Rightarrow x=2y-13$	M1
	Solves simultaneous equations;	
	$180=([2y-13]+1)^2+(y-6)^2\Rightarrow 0=5y^2-60y$	M1M1
	or $180=(x+1)^2+\left(\frac{1}{2}x+\frac{13}{2}-6\right)^2\Rightarrow 0=x^2+2x-143=0$	
	$y=0,\ y=12\Rightarrow x=-13,\ x=11$ or $x=-13,\ x=11\Rightarrow y=0,\ y=12$	A1A1
	Coordinates are $(-13,\ 0)$ and $(11,\ 12)$	[8]
(c)	Area of triangle PQR	
	$PQ=\sqrt{(6+2)^2+(-1-3)^2}=4\sqrt{5}$	M1
	Area $=\frac{1}{2}\times 4\sqrt{5}\times 6\sqrt{5}=60$ (units)^2	M1A1 [3]
	ALT	
	Area $=\frac{1}{2}\begin{pmatrix}-13 & -1 & 3 & -13\\ 0 & 6 & -2 & 0\end{pmatrix}=\frac{1}{2}(-78+2+0-0-18-26)=-60$	{M1}{M1}
	$\Rightarrow 60$ (units)^2	{A1} [3]
(d)	Coordinates of R required are $(-13,\ 0)$	
	$\angle RPQ=90^\circ$ so RQ is a diameter	
	$\left(\frac{-13+3}{2},\frac{0-2}{2}\right)\Rightarrow(-5,\ -1)$	M1A1 [2]
		Total 15 marks

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(b)	
B1	Gradient of $l_1 = -2$
B1	Gradient of $l_2 = \frac{1}{2}$
M1	Use of $PR = 6\sqrt{5}$ to obtain an equation
M1	Use of gradient of the perpendicular to obtain an equation
M1	Solves simultaneously
M1	Simplifies to $5y^2 - 60y = 0$ or $x^2 + 2x - 143 = 0$
A1	All 4 values identified 0, 12, –13, 11
A1	$(11,\ 12)$ and $(-13,\ 0)$ (must be paired correctly and if written as a coordinate then must be in the correct order)
(c)	
M1	$PQ = 4\sqrt{5}$
M1	Use of Area $= \frac{1}{2} \times PQ \times PR$
A1	60 (units)^2
ALT	
M1	Use of Area $= \frac{1}{2}\begin{pmatrix} -13 & -1 & 3 & -13 \\ 0 & 6 & -2 & 0 \end{pmatrix}$ ft R provided $e < 0$
M1	$\frac{1}{2}(-78 + 2 + 0 - 0 - 18 - 26)$ ft R provided $e < 0$
A1	60 (units)^2
(d)	
M1	$\left(\frac{-13+3}{2}, \frac{0-2}{2}\right)$ ft R provided $e < 0$
A1	$(-5,\ -1)$

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Question number	Scheme	Marks

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Answer 6 of 10 · Question 4 part b · 2 marks

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number		
4 (a)	Gradient $= \dfrac{11+10}{3+4} = 3$	
	$y+10=3(x+4)$ or $y-11=3(x-3)$ oe	M1A1 (2)
(b)	e.g. $\left(\dfrac{4\times-4+3\times3}{3+4}, \dfrac{4\times-10+3\times11}{3+4}\right)=(-1,-1)$	M1 A1 (2)
ALT (b)	Using Vectors	
	$\begin{pmatrix}-4\\-10\end{pmatrix}+\dfrac{3}{7}\begin{pmatrix}7\\21\end{pmatrix}=\begin{pmatrix}-1\\-1\end{pmatrix}$ or $\begin{pmatrix}3\\11\end{pmatrix}-\dfrac{4}{7}\begin{pmatrix}7\\21\end{pmatrix}=\begin{pmatrix}-1\\-1\end{pmatrix}$	{M1} {A1}
(c)	$-\dfrac{1}{3}=\dfrac{n+1}{m+1} \Rightarrow -\dfrac{1}{3}(m+1)=n+1$	M1
	$\left(\sqrt{10}\right)^2=(m+1)^2+(n+1)^2$	M1
	$10=(m+1)^2+\dfrac{1}{9}(m+1)^2$	M1
	$9=(m+1)^2$	M1
	$m=-4 \quad n=0$	A1 A1 (6)
ALT (c)	Using Vectors	
	$\overrightarrow{AB}=\begin{pmatrix}7\\21\end{pmatrix}$ so perpendicular to $\overrightarrow{AB}=\begin{pmatrix}21\\-7\end{pmatrix}$	{M1}
	$\left\lvert\overrightarrow{AB}\right\rvert=7\sqrt{10} \Rightarrow, \left\lvert\overrightarrow{AP}\right\rvert=3\sqrt{10}$	{M1,M1}
	$\overrightarrow{PQ}=\dfrac{\sqrt{10}}{7\sqrt{10}}\times\begin{pmatrix}21\\-7\end{pmatrix}=\begin{pmatrix}3\\-1\end{pmatrix}$	{M1}
	So $Q=(-1-3,-1--1)$	{A1}
	$Q=(-4,0)$	{A1}
(d)(i)	$AB=\sqrt{(3+4)^2+(11+10)^2}=7\sqrt{10}$	M1
	$RQ=\sqrt{(-11+4)^2+(-21)^2}=7\sqrt{10}$	A1
(d)(ii)	Gradient of $RQ=\dfrac{-21-0}{-11+4}=3$	M1
	So Gradient of AB (=3) = Gradient of RQ	A1 (4)

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Answer 6 of 10 · Question 4 part b · 2 marks

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ALT (d)	Using Vectors $\overrightarrow{RQ} = \begin{pmatrix} -4-(-11) \\ 0-(-21) \end{pmatrix} = \begin{pmatrix} 7 \\ 21 \end{pmatrix}$	{M1} {A1}
	$\overrightarrow{AB} = \begin{pmatrix} 7 \\ 21 \end{pmatrix} = \overrightarrow{RQ}$	{M1}
	Because the vectors are the same they must be parallel and the same length	{A1}
(e)	Area = $7\sqrt{10} \times \sqrt{10} = 70$	M1 A1 (2)
ALT (e)	Using Vectors $\frac{1}{2}\begin{vmatrix} 3 & -4 & -11 & -4 & 3 \\ 11 & 10 & -21 & 0 & 11 \end{vmatrix}$	{M1}
	$= 70$	{A1}
		Total is 16 marks

Type MS · Midpoints and points that split a line in a ratio

Answer 6 of 10 · Question 4 part b · 2 marks

2021 JUN · Paper 02 · extract 3 of 5 · find part b in this mark scheme extract

Part	Mark	Guidance
(a)	M1	For a fully correct method of finding an equation of a straight line. $\frac{y-y_1}{y_2-y_1}=\frac{x-x_1}{x_2-x_1}\Rightarrow\frac{y-(-10)}{11-(-10)}=\frac{x-(-4)}{3-(-4)}$ Or finds gradient $\frac{11+10}{3+4}=3$ and uses $y+10=3(x+4)$ or $y-11=3(x-3)$ If $y=mx+c$ is used, they must find a complete equation for this mark. Allow **one** error only for the award of this mark.
	A1	For a correct line in any form. $y+10=3(x+4)$ or $y-11=3(x-3)$ or $y=3x+2$ or even $\frac{y+10}{21}=\frac{x+4}{7}$ but do not allow incomplete processing.
(b)	M1	For one correct from $x=-1$ or $y=-1$
	A1	For the correct coordinates of point P $(-1,-1)$ Accept $x=-1$ $y=-1$
(c)	M1	Uses the perpendicular gradient to set up an equation in m and n. $-\frac{1}{'3'}=\frac{n-'(-1)'}{m-'(-1)'}\Rightarrow-\frac{1}{'3'}(m+1)=n+1$ or $n=-\frac{1}{3}m-\frac{4}{3}$ Ft their gradient in part (a) and their P from part (b) for this mark.
	M1	Uses Pythagoras theorem to set up an equation in m and n. $\left(\sqrt{10}\right)^2=(m-'(-1)')^2+(n-'(-1)')^2$ Ft their coordinates of point P form part (b) for this mark.
	M1	Attempts to solve their two equations in n and m simultaneously and forms a quadratic equation in one variable only. $10=(m+1)^2+\frac{1}{9}(m+1)^2\Rightarrow 9=(m+1)^2$ or $0=m^2+2m-8$ or $10=9(n+1)^2+(n+1)^2\Rightarrow 0=10n^2+20n$
	M1	For solving their either: $9=(m+1)^2\Rightarrow m=...$ or $0=10n^2+20n\Rightarrow n=...$ which must be a quadratic equation.
	A1	For finding **either** $m=-4$ **or** $n=0$ Condone the sight of $m=2$ for this mark.
	A1	For finding **both** $m=-4$ **and** $n=0\Rightarrow(-4,\ 0)$ The final answer must be given as coordinates.
	ALT – using vectors – see main scheme.	
(d)(i)	M1	For finding **either** the length $AB=\sqrt{(3+4)^2+(11+10)^2}=7\sqrt{10}$ **Or** $RQ=\sqrt{(-11+4)^2+(-21)^2}=7\sqrt{10}$
	A1	For finding **both** the length $AB=\sqrt{(3+4)^2+(11+10)^2}=7\sqrt{10}$ **And** $RQ=\sqrt{(-11+4)^2+(-21)^2}=7\sqrt{10}$ **and** states they are equal
(d)(ii)	M1	The gradient of $RQ=\frac{-21-'0'}{-11-'(-4)'}='3'$ Ft their coordinates from part (c)

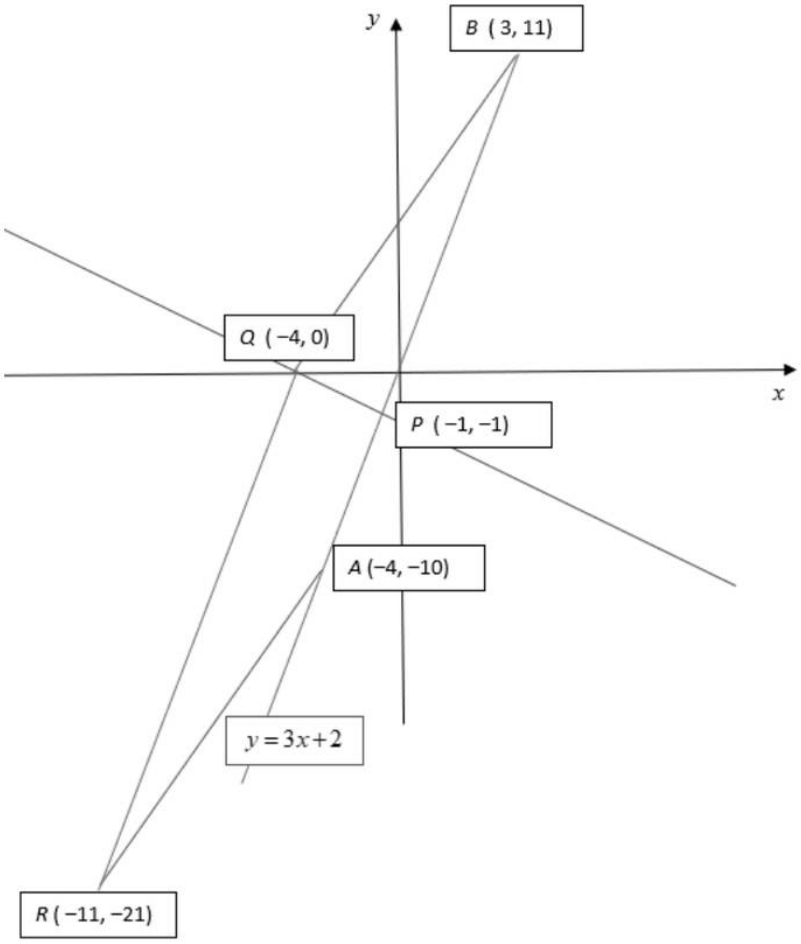
Type MS · Midpoints and points that split a line in a ratio

Answer 6 of 10 · Question 4 part b · 2 marks

2021 JUN · Paper 02 · extract 4 of 5 · find part b in this mark scheme extract

	A1	States that the gradient of RQ = gradient of AB [from (a)]
	ALT – Uses vectors, see main scheme. Ft their coordinates of Q – (m, n)	
(e)	**M1**	For a correct expression for the area using their length of AB and the given length of PQ $\left(\sqrt{10}\right)$ Area = $'7\sqrt{10}'\times\sqrt{10}=...$
	A1	For the area = 70 [square units]
	ALT – Uses the discriminant	
	M1	For a correct expression of the area in sequential order using their coordinates for Q Area = $\frac{1}{2}\begin{vmatrix} 3 & -4 & -11 & '-4' & 3 \\ 11 & 10 & -21 & '0' & 11 \end{vmatrix}$
	A1	Area = 70 [square units]

Useful sketch



Type MS · Midpoints and points that split a line in a ratio

Answer 6 of 10 · Question 4 part b · 2 marks

2021 JUN · Paper 02 · extract 5 of 5 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Midpoints and points that split a line in a ratio

Answer 7 of 10 · Question 8 part d · 3 marks

2023 JUN · Paper 02 (regional R) · extract 1 of 5 · find part d in this mark scheme extract

8(a)	$\frac{y-5}{5-9}=\frac{x-1}{1-9}$ $\Rightarrow x-2y+9=0$	M1A1 A1 [3]
(b)	Coordinates of point X $\left(\frac{3\times9+1\times1}{3+1},\frac{3\times9+1\times5}{3+1}\right)=(7,8)$ The perpendicular gradient $=-\frac{1}{\frac{1}{2}}=-2$ Equation of l $y-8=-2(x-7)\Rightarrow y=-2x+22$ *	B1B1 B1ft M1A1 cso [5]
(c)	$y=-2(6)+22\Rightarrow p=10$	B1 [1]
(d)	$\overrightarrow{BA}=\begin{pmatrix}1-9\\5-9\end{pmatrix}=\begin{pmatrix}-8\\-4\end{pmatrix}\Rightarrow \overrightarrow{CD}=\begin{pmatrix}6-8\\10-4\end{pmatrix}=\begin{pmatrix}-2\\6\end{pmatrix}$ and coordinates of C are (6, 10) so coordinates of D are $(-2, 6)$	M1 A1A1 [3]
(e)	Length of AB (or CD) $=\sqrt{(9-5)^2+(9-1)^2}=\sqrt{80}$ Length of $CX=\sqrt{(10-8)^2+(7-6)^2}=\sqrt{5}$ Area of parallelogram $ABCD=\sqrt{5}\times\sqrt{80}=\sqrt{400}=20$ (units^2)	B1 B1 M1A1 [4]
		Total 16 marks

Part	Mark	Notes
(a)	M1	For using a correct method and the given coordinates of A and B to form the equation of AB. Do not score this mark until they find the gradient and form the equation of the line using a correct formula. If they use $y=mx+c$ do not allow this mark until they find c and form a complete equation. $\left[m=\frac{1}{2},\quad c=\frac{9}{2}\quad y=\frac{x}{2}+\frac{9}{2}\right]$
	A1	For the correct equation of AB in any form.
	A1	For the correct equation of AB in the required form.

Type MS · Midpoints and points that split a line in a ratio

Answer 7 of 10 · Question 8 part d · 3 marks

2023 JUN · Paper 02 (regional R) · extract 2 of 5 · find part d in this mark scheme extract

(b)	B1	For either x or y correct coordinates of point X **NB This is an M mark in Epen**
	B1	For both x and y correct coordinates of point X **NB This is an A mark in Epen.**
	B1ft	For writing down the inverse reciprocal of their gradient for AB Ft their gradient form part (a)
	M1	For forming an equation for l using their coordinates of X and their negative reciprocal gradient of AB $y-8=-2(x-7)\Rightarrow y=-2x+22$
	A1	For the correct equation of l as shown.
(c)	B1	For $y = 10$
(d)	M1	For a suitable method. **Method 1 - Uses vectors:** $\overrightarrow{BA}=\begin{pmatrix}1-9\\5-9\end{pmatrix}=\begin{pmatrix}-8\\-4\end{pmatrix}\Rightarrow\overrightarrow{CD}=\begin{pmatrix}6+(-8)\\10+(-4)\end{pmatrix}\Rightarrow\left[\begin{pmatrix}-2\\6\end{pmatrix}\right]$ OR: $\overrightarrow{BC}=\begin{pmatrix}6-9\\9-10\end{pmatrix}=\begin{pmatrix}-3\\-1\end{pmatrix}\Rightarrow\overrightarrow{AD}=\begin{pmatrix}1-3\\5-(-1)\end{pmatrix}\Rightarrow\left[\begin{pmatrix}-2\\6\end{pmatrix}\right]$ **Method 2 - Uses simultaneous equations**: The equation of AD is $y=-\frac{x}{3}+\frac{16}{3}$ and of CD is $y=\frac{x}{2}+7$ $-\frac{x}{3}+\frac{16}{3}=\frac{x}{2}+7\Rightarrow x=-2$ and $y=6$ **Method 3 - Uses the gradient and length of *AB* or *CD*** $AB=CD=4\sqrt{5}=\sqrt{(6-x)^2+(y-10)^2}$ Gradient: $\frac{1}{2}=\frac{10-y}{6-x}\Rightarrow x=2y-14$ $\left(4\sqrt{5}\right)^2=(6-(2y-14))x^2+(y-10)^2\Rightarrow 5y^2-100y+420=0$ $\Rightarrow y=6,14$ and $x=-2,14$ Allow no more than one error in either method.
	A1	For either correct x or y coordinate of D [(– 2, 6)] **NB** this is an A mark in Epen
	A1	For both correct coordinates of D (– 2, 6)

Type MS · Midpoints and points that split a line in a ratio

Answer 7 of 10 · Question 8 part d · 3 marks

2023 JUN · Paper 02 (regional R) · extract 3 of 5 · find part d in this mark scheme extract

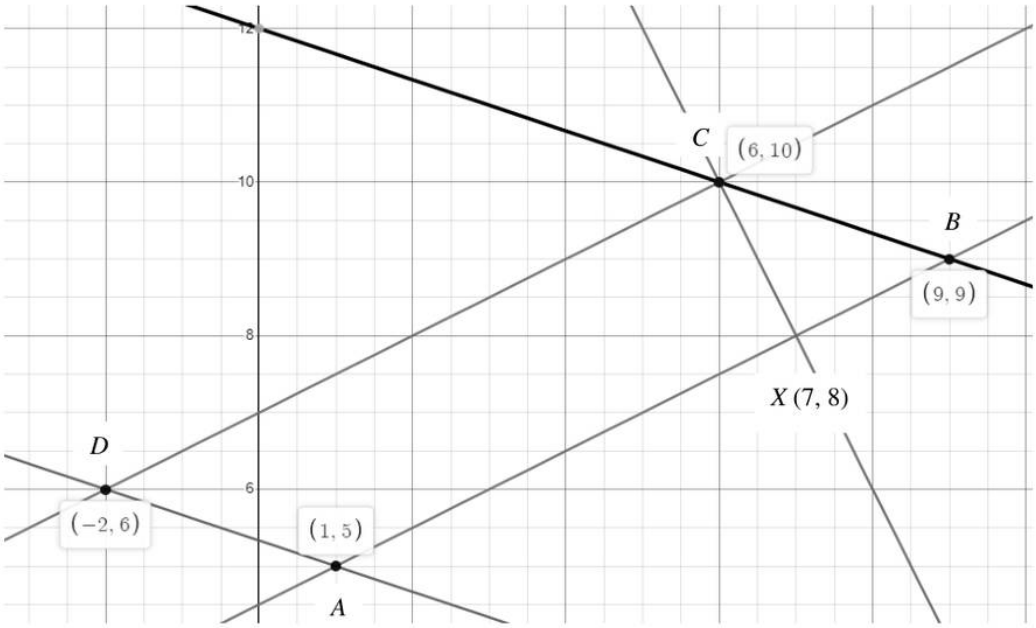
(e)	B1	For the correct length of either ***AB* [*CD*] or *CX*** These are given coordinates there is no ft
	B1	For the correct lengths of both ***AB* [*CD*] and *CX*** Ft their C and their X
	M1	For using a correct method to calculate the area of a parallelogram. $\text{b} \times \text{h} = CX \times AB = \sqrt{5} \times \sqrt{80} = \sqrt{400} = 20$ If their lengths are incorrect allow this mark provided their lengths are identified as base and perpendicular height. If they use AD as the base $\left[\sqrt{10}\right]$, the perpendicular height required is $2\sqrt{10}$ so base × height = $\sqrt{10} \times 2\sqrt{10} = ...$
	A1	For the correct area of 20 [units^2]
	ALT – Using determinants	
	B1B1	For sight of the correct array which must use the coordinates of A, B, C and D **ONLY.** The coordinates (7, 8) seen in the array is B0B0 $\begin{pmatrix} 1 & 9 & '6' & '-2' & 1 \\ 5 & 9 & '10' & '6' & 5 \end{pmatrix}$ Award both marks for fully correct. Award B1 if there are no more than 2 errors but with none missing. There must be 5 sets of coordinates in the array with first and last the same. **The coordinates must go in order around the parallelogram clockwise or anticlockwise.**
	M1	For the correct evaluation of their 2×5 array. Allow this even if they have (7, 8) included instead of (9, 9) [which is a common error] $\frac{1}{2}\begin{pmatrix} 1 & 9 & 6 & -2 & 1 \\ 5 & 9 & 10 & 6 & 5 \end{pmatrix} = \frac{1}{2}\left\lvert\left[(9+90+36-10)-(45+54-20+6)\right]\right\rvert = ...$
	A1	For the correct area of 20 [units^2]

Type MS · Midpoints and points that split a line in a ratio

Answer 7 of 10 · Question 8 part d · 3 marks

2023 JUN · Paper 02 (regional R) · extract 4 of 5 · find part d in this mark scheme extract

Useful sketch



Type MS · Midpoints and points that split a line in a ratio

Answer 7 of 10 · Question 8 part d · 3 marks

2023 JUN · Paper 02 (regional R) · extract 5 of 5 · find part d in this mark scheme extract

Question	Scheme	Marks

Type MS · Midpoints and points that split a line in a ratio

Answer 8 of 10 · Question 11 part c · 2 marks

2021 NOV · Paper 02 · extract 1 of 3 · find part c in this mark scheme extract

11 (a)	$\frac{y-5}{5--5}=\frac{x-7}{7--3}$ $[\Rightarrow y=x-2]$ oe	M1A1 [2]
(b)	$m_{BC}=\frac{5-1}{7-p}=\frac{4}{7-p}$, $m_{AC}=\frac{1-(-5)}{p-(-3)}=\frac{6}{p+3}$ $m_{BC}\times m_{AC}=-1\Rightarrow\frac{4}{7-p}\times\frac{6}{p+3}=-1\Rightarrow\frac{4}{7-p}=-\left(\frac{p+3}{6}\right)$ $\Rightarrow -24=-p^2+4p+21\Rightarrow p^2-4p-45=0$ $\Rightarrow (p-9)(p+5)=0\Rightarrow p=-5,\ (9)$	M1A1A1 M1 dM1 M1A1* [7]
(c)	Using formula; $\left(\frac{(7\times-5)-(3\times7)}{7-3},\frac{(7\times1)-(3\times5)}{7-3}\right)=(-14,-2)$	B1B1 [2]
(d) (i)	$AC=\sqrt{(1-[-5])^2+(-5-[-3])^2}=\sqrt{40}=2\sqrt{10}$	B1
(ii)	$BD=\sqrt{(7-[-14])^2+(5-[-2])^2}=\sqrt{490}=7\sqrt{10}$ Area $=\frac{1}{2}\times2\sqrt{10}\times7\sqrt{10}=70\ (\text{units}^2)$	M1 M1A1 [4]
		Total 15 marks

Type MS · Midpoints and points that split a line in a ratio

Answer 8 of 10 · Question 11 part c · 2 marks

2021 NOV · Paper 02 · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Notes
(a)	M1	For forming an equation of a straight line
	A1	Accept a correct line in any form following simplification of denominators; for example, accept $\frac{y-5}{10}=\frac{x-7}{10}$ or $y-5=x-7$ or $y+5=x+3$
(b)	M1	For attempting to find an expression for the gradient of BC **or** the gradient of AC $m_{BC}=\frac{5-1}{7-p}$ or $m_{AC}=\frac{1-(-5)}{p-(-3)}$ Accept maximum one slip in either expression for this mark.
	A1	For an expression for the gradient of BC **or** AC fully correct
	A1	For an expression for the gradient of BC **and** AC fully correct Need not be simplified in either case
	M1	For setting their gradient of BC = negative reciprocal of their gradient of AC
	dM1	For attempting to form a 3TQ in p: $\left[p^2-4p-45=0\right]$ Dependent on the previous M mark.
	M1	For solving their 3TQ by any method. See General Guidance.
	A1*	For $p=-5$
(c)	B1	For either $x=-14$ or $y=-2$
	B1	Both coordinates correct $(-14,-2)$ given as coordinates only.
(d)(i)	B1	For the correct length of AC $\left[=2\sqrt{10}\right]$
(ii)	M1	For the correct method to find length BD $\left[=7\sqrt{10}\right]$ Ft their $x=-14$ or $y=-2$
	M1	For a correct method using $A=\frac{1}{2}\times\text{base}\times\text{height}$ using their AC and BD
	A1	For the correct area of $70\ (\text{units}^2)$
	ALT – Uses determinants	
	M1	For stating $A=\frac{1}{2}\begin{pmatrix}-3 & 7 & -14 & -3\\ -5 & 5 & -2 & -5\end{pmatrix}$ oe
	M1	For evaluating their determinant $A=\frac{1}{2}\left(\left[-3\times5+7\times-2+-14\times-5\right]-\left[7\times-5+-14\times5+-3\times-2\right]\right)$ oe
	A1	Area = $70\ (\text{units}^2)$ Which must be a positive value

Type MS · Midpoints and points that split a line in a ratio

Answer 8 of 10 · Question 11 part c · 2 marks

2021 NOV · Paper 02 · extract 3 of 3 · find part c in this mark scheme extract

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Type MS · Midpoints and points that split a line in a ratio

Answer 9 of 10 · Question 8 part c · 2 marks

2023 JUN · Paper 02 · extract 1 of 5 · find part c in this mark scheme extract

Question	Scheme	Marks
8(a)	$(\text{Grad}_{AB}) = \frac{2-8}{12-(-6)}$ oe $y-2="\left(-\frac{1}{3}\right)"(x-12)$ oe $\Rightarrow x+3y-18=0$ or $-x-3y+18=0$ oe	M1 dM1 (A1 on ePen) A1 [3]
(b)	Length $=\sqrt{(12--6)^2+(2-8)^2}=6\sqrt{10}$ eg $\sqrt{360}$ oe	M1A1 [2]
(c)	$\left(\left(\frac{2\times-6+1\times12}{1+2}, \frac{2\times8+1\times2}{1+2}\right)\right)$ $(0,6)$ or $m=0, n=6$	B1 B1 (M1A1 on ePen) [2]
(d) Mark parts (i) and (ii) together	Gradient of $CA=\frac{q-8}{p+6}$ **or** gradient of $CB=\frac{q-2}{p-12}$ oe $\frac{q-8}{p+6}=-\frac{1}{\frac{q-2}{p-12}}=-\left(\frac{p-12}{q-2}\right)\Rightarrow q^2-10q+16=-p^2+6p+72$ Gradient of $XC=-\frac{1}{"-\frac{1}{3}"}\Rightarrow-\frac{1}{"-\frac{1}{3}"}=\frac{q-6}{p-0}(\Rightarrow q=3p+6)$ $('3p+6')^2-10('3p+6')+16=-p^2+6p+72\Rightarrow 10p^2-80=0$ $10p^2-80=0\Rightarrow p=\sqrt{8}$ oe $q=3\times\sqrt{8}+6=6+6\sqrt{2}$ oe	M1 M1 B1ft ddM1A1 M1A1 [7]
ALT1 Mark parts (i) and (ii) together	$(\text{midpoint of } AB=)\left(\frac{-6+12}{2}, \frac{8+2}{2}\right)$ $(=(3,5))$ $(\text{radius of } C)=\frac{"6\sqrt{10}"}{2}$ oe or $\sqrt{(-6-3)^2+(8-5)^2}$ oe $y=\frac{-1}{"-\frac{1}{3}"}x+6$ $(y=3x+6)$ oe $(x-"3")^2+(y-"5")^2=\left(\frac{"6\sqrt{10}"}{2}\right)^2\Rightarrow(x-"3")^2+("3x+6"-"5")^2=\left(\frac{"6\sqrt{10}"}{2}\right)^2$ $10x^2=80$ oe $x=\sqrt{8}$ oe $p=\sqrt{8}, q=6+3\sqrt{8}$ oe	M1 M1 B1ft ddM1 A1 M1 A1 [7]

Type MS · Midpoints and points that split a line in a ratio

Answer 9 of 10 · Question 8 part c · 2 marks

2023 JUN · Paper 02 · extract 2 of 5 · find part c in this mark scheme extract

ALT2 Mark parts (i) and (ii) together	$\left((AC)^2=\right)(p--6)^2+(q-8)^2$ or $(AC=)\sqrt{(p--6)^2+(q-8)^2}$ oe	M1
	$\left((BC)^2=\right)(p-12)^2+(q-2)^2$ or $(AC=)\sqrt{(p-12)^2+(q-2)^2}$ oe	M1
	$q=3p+6$	B1ft
	$\left((AB)^2=(AC)^2+(BC)^2\right)$	
	$\left("6\sqrt{10}"\right)^2=(p--6)^2+("3p+6"-8)^2+(p-12)^2+("3p+6"-2)^2$	ddM1A1
	$10p^2=80$ oe	M1
	$p=\sqrt{8}$ oe	A1
	$p=\sqrt{8}, q=6+3\sqrt{8}$ oe	[7]
(e)	Length $CX=\sqrt{\left("\sqrt{8}"-"0"\right)^2+\left("6+6\sqrt{2}"-"6"\right)^2}\left(=4\sqrt{5}\right)$	M1
	Area of triangle $ABC=\frac{1}{2}\times"4\sqrt{5}"\times"6\sqrt{10}"=60\sqrt{2}$ oe	dM1 A1 [3]
ALT	$\frac{1}{2}\begin{vmatrix}-6 & 12 & "\sqrt{8}" & -6\\ 8 & 2 & "6+3\sqrt{8}" & 8\end{vmatrix}$ oe	M1
	$\frac{1}{2}\left[\left(-6\times2+12\times"(6+3\sqrt{8})"+"\sqrt{8}"\times8\right)-\left(-6\times"(6+3\sqrt{8})"+"\sqrt{8}"\times2+12\times8\right)\right]$ oe	dM1
	$60\sqrt{2}$ oe	A1 [3]
		Total 17 marks

Type MS · Midpoints and points that split a line in a ratio

Answer 9 of 10 · Question 8 part c · 2 marks

2023 JUN · Paper 02 · extract 3 of 5 · find part c in this mark scheme extract

Part	Mark	Notes
(a)	M1	For correctly finding the gradient of AB in unsimplified form.
	dM1 (A1 on ePen)	For a full and correct attempt to find the equation of the line using their gradient. No simplification is required. If using $y = mx + c$, a value for c must be found.
	A1	For a correct equation in the required form.
(b)	M1	For using a correct method to find the length of line segment AB, in unsimplified form.
	A1	For the correct exact length.
(c)	B1 (M1 on ePen)	For either coordinate correct.
	B1 (A1 on ePen)	For both correct coordinates (0, 6) For part c the values of m and n can be explicitly identified or written in a coordinate.
In part (d) allow p to be interchangeable with x, q to be interchangeable with y throughout		
(d)	M1	For a correct statement of the gradient for either CA or CB
	M1	For using the negative perpendicular of either gradient and equating the gradients to form an equation in terms of p and q only.
	B1ft	For finding the negative reciprocal of their gradient of XC and placing this equal to a correct expression in p and q as shown.
	ddM1	For correctly substituting their **linear** expression for p or q into a **quadratic** equation in q or p to obtain an equation in one variable. Must use their $q = 3p + 6$ and is dependent on both previous method marks.
	A1	For the correct two term quadratic..
	M1	For **correctly solving their** quadratic to find a value for either p or q
	A1	For **both** p and q correct.
ALT1 (d)	M1	For the correct method to find the midpoint of AB
	M1	For the correct method to find the radius of C, ft their answer from part b if used.
	B1ft	For the equation of the line, unsimplified, ft their gradient of AB
	ddM1	For correctly substituting their $y = 3x + 6$ into the equation of a circle, using their midpoint of AB and their radius of C Must use their $y = 3x + 6$ and is dependent on both previous method marks.
	A1	Correct equation
	M1	For **correctly solving their** quadratic to find a value for either x or y
	A1	For **both** p and q correct.
ALT2 (d)	M1	For the correct method to find the length of AC or $(AC)^2$
	M1	For the correct method to find the length of BC or $(BC)^2$
	B1ft	For the equation of the line, unsimplified, ft their gradient of AB
	ddM1	For correctly substituting their lengths and their $q = 3p + 6$ into a correct Pythagorean equation, dependent on both previous method marks.
	A1	Correct equation
	M1	For **correctly solving their** quadratic to find a value for either p or q
	A1	For **both** p and q correct.

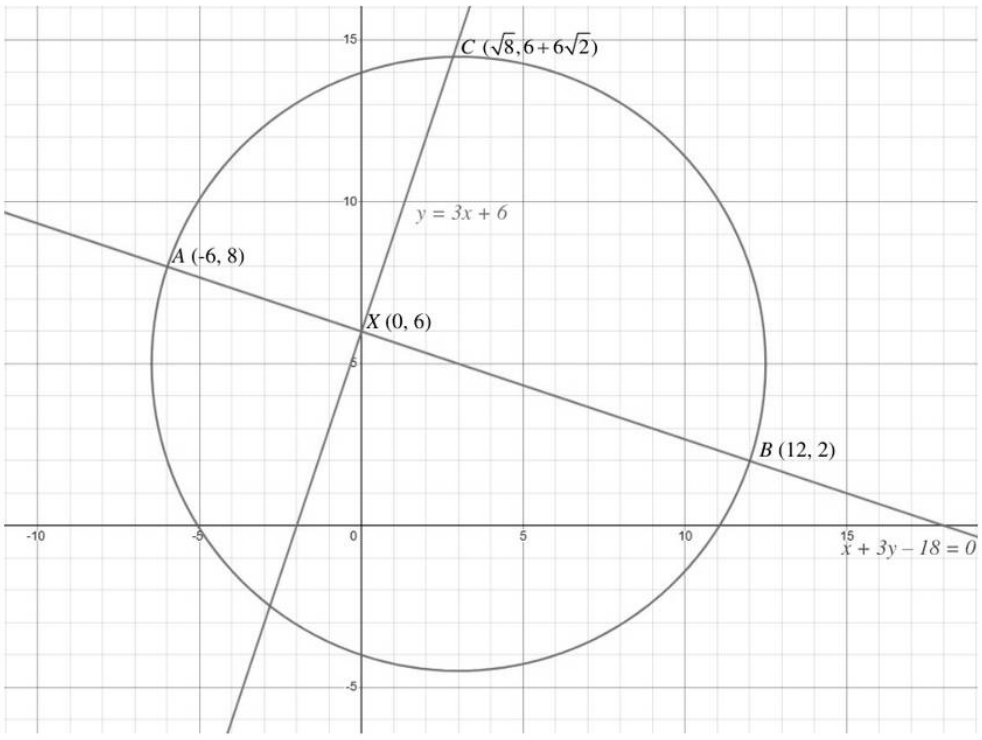
Type MS · Midpoints and points that split a line in a ratio

Answer 9 of 10 · Question 8 part c · 2 marks

2023 JUN · Paper 02 · extract 4 of 5 · find part c in this mark scheme extract

(e)	**M1**	For using a correct method to find the length of the perpendicular from AB to C
	dM1	For using their results from part (b) and their length of perpendicular from AB to C with the correct formula for the area of a triangle. Dependent on previous method mark.
	A1	For the correct area of $60\sqrt{2}$ or $30\sqrt{8}$ (units^2)
ALT	**M1**	For a correct statement for the area such as the one shown, using their values of p and q
	dM1	For the correct evaluation of their determinant Dependent on previous method mark.
	A1	For the correct area of $60\sqrt{2}$ or $30\sqrt{8}$ (units^2)

Useful Sketch for Parts c/d – look for any working on or near a sketch.



Type MS · Midpoints and points that split a line in a ratio

Answer 9 of 10 · Question 8 part c · 2 marks

2023 JUN · Paper 02 · extract 5 of 5 · find part c in this mark scheme extract

Question	Scheme	Marks

Type MS · Midpoints and points that split a line in a ratio

Answer 10 of 10 · Question 2 part c · 2 marks

2023 JAN · Paper 01 · extract 1 of 5 · find part c in this mark scheme extract

2 a	(Gradient of AB =) $\frac{0-3}{4+5}\left(=-\frac{1}{3}\right)$	M1
	Gradient of the perpendicular = $-\frac{1}{"-\frac{1}{3}"}(=3)$	M1
	$y-5=-\frac{1}{"-\frac{1}{3}"}(x+1)$ or $y-5="3"(x+1)$ or $y=3x+8$	M1
	$3x-y+8=0$ or $-3x+y-8=0$ oe	A1 (4)
b	(Equation of AB =) $y-0="-\frac{1}{3}"(x-4)$ or $y=-\frac{1}{3}x+\frac{4}{3}$ oe	M1
	Solve $y-3x-8=0$ and $y-0=-\frac{1}{3}(x-4)$ oe simultaneously	M1
	$0=\frac{10}{3}x+\frac{20}{3}$ oe	
	$x=-2$ $y=2$ or (-2, 2) *	A1 cso (3)
	Alternative for final 2 marks (substitutes (- 2, 2) into each equation))	
	Line l is $3x-y+8=0$	
	When $x=-2$ and $y=2$, $-6-2+8=0\Rightarrow D$ lies on l	{M1}
	AB is $y=-\frac{1}{3}\ x-4$	
	When $x=-2$ and $y=2$, $-\frac{1}{3}\ -2-4\ =2=y\Rightarrow D$ lies on AB	
	So l and AB intersect at $-2, 2$ *	{A1 cso} (3)
c	Midpoint of $AB=\left(\frac{-5+4}{2},\frac{3(+0)}{2}\right)=\left(-\frac{1}{2},\frac{3}{2}\right)$	B1 (M1 on ePen)
	$\neq(-2,2)$ So not perpendicular bisector of AB *	B1 (A1 on ePen) cso (2)
ALT	$BD=\sqrt{(4--2)^2+(0-2)^2}\left(=\sqrt{40}\right)$ or $AD=\sqrt{(-5--2)^2+(3-2)^2}\left(=\sqrt{10}\right)$	M1
	$BD=\sqrt{40}(=2\sqrt{10})$ and $AD=\sqrt{10}$	A1cso
	$BD\neq AD$ so not the perpendicular bisector	(2)

Type MS · Midpoints and points that split a line in a ratio

Answer 10 of 10 · Question 2 part c · 2 marks

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d	$BD = \sqrt{(4--2)^2+(0-2)^2}\ (=\sqrt{40})$ oe	M1
	$CD = \sqrt{(-1--2)^2+(5-2)^2}\ (=\sqrt{10})$ oe	M1
	$(\tan\angle ABC =)\ \dfrac{"\sqrt{(-1--2)^2+(5-2)^2}"}{"\sqrt{(4--6)^2+(0-2)^2}"}$ or $\dfrac{\sqrt{10}}{\sqrt{40}}$	ddM1
	$=\dfrac{1}{2}$	A1 (4)
ALT 1	Uses cosine rule in triangle ABC	
	$AB = \sqrt{(-5-4)^2+(3-0)^2}\ (=\sqrt{90}=3\sqrt{10})$ or $BC = \sqrt{(4--1)^2+(0-5)^2}\ (=\sqrt{50}=5\sqrt{2})$ or $AC = \sqrt{(-1--5)^2+(5-3)^2}\ (=\sqrt{20}=2\sqrt{5})$	M1
	may also find expressions for $(AB)^2, (BC)^2$ or $(AC)^2$	
	$AB = \sqrt{(-5-4)^2+(3-0)^2}\ (=\sqrt{90}=3\sqrt{10})$ and $BC = \sqrt{(4--1)^2+(0-5)^2}\ (=\sqrt{50}=5\sqrt{2})$ and $AC = \sqrt{(-1--5)^2+(5-3)^2}\ (=\sqrt{20}=2\sqrt{5})$	M1
	may also find expressions for $(AB)^2, (BC)^2$ and $(AC)^2$	
	$(\cos ABC =)\ \dfrac{("\sqrt{50}")^2+("\sqrt{90}")^2-("\sqrt{20}")^2}{2\times"\sqrt{50}"\times"\sqrt{90}"}\left(=\dfrac{2}{\sqrt{5}}=\dfrac{2\sqrt{5}}{5}\right)$	ddM1
	$(\tan ABC)=\dfrac{1}{2}$	A1
		Total 13 marks

Type MS · Midpoints and points that split a line in a ratio

Answer 10 of 10 · Question 2 part c · 2 marks

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Part	Mark	Additional Guidance
(a)	M1	For correctly finding the gradient of AB, need not be simplified.
	M1	For finding the negative reciprocal of their gradient of AB, need not be simplified.
	M1	For a fully correct method using their negative reciprocal to find the equation of the straight line. Allow use of their $-\frac{1}{"-\frac{1}{3}"}$ which can be unsimplified or processed incorrectly, but must follow from a clear attempt to have found the negative reciprocal of their gradient for AB. If $y = mx + c$ is used, there must be a fully correct substitution and a fully correct rearrangement to find c
	A1	For $3x - y + 8 = 0$ or $-3x + y - 8 = 0$ oe so long as integer coefficients and = 0.
(b)	M1	For writing the equation of AB, the equation need not be simplified, but must be fully correct.
	M1	For an attempt to solve their equations simultaneously, allow one processing error. Must eliminate one variable and arrive at a value for x or y.
	A1 cso*	For obtaining the required result, no errors.
	ALT	**Final 2 marks**
	M1	For writing the equation of AB, the equation need not be simplified, but must be fully correct.
	M1	For a fully correct substitution of (–2, 2) into a correct equation for AB to show that D lies on AB **OR** a fully correct substitution of (–2, 2) into **their** equation for line l
	A1 cso*	For a fully correct substitution of (–2, 2) into a correct equation for AB to show that D lies on l **AND** a fully correct substitution of (–2, 2) into a correct equation for line l to show that D lies on l. An appropriate (minimal) conclusion must also be written, this can be as simple as # or 'shown'. No errors.
(c)	B1 (M1 on ePen)	For either coordinate of the midpoint correct
	B1 (A1 on ePen) cso*	For both coordinates correct and an appropriate (minimal) conclusion, which must be correct, can be as simple as # or 'shown'.
ALT	M1	For a correct method to calculate AD or BD
	A1 cso*	For correct values for AD and BD, accept decimals and a (minimal) conclusion, can be as simple as # or 'shown'.
(d)	M1	For the correct method to calculate BD (May be seen in (c))
	M1	For the correct method to calculate CD
	ddM1	For $\left(\tan\angle ABC =\right)\frac{\sqrt{(-1--2)^2+(5-2)^2}}{\sqrt{(4--6)^2+(0-2)^2}}$ or $\frac{\sqrt{10}}{\sqrt{40}}$ allow their BD and CD. Dependent on the previous 2 method marks.
	A1	For $\left(\tan\angle ABC\right)=\frac{1}{2}$
If you see a method using the gradient of BC *and* tan $(A - B)$ – PLEASE SEND TO REVIEW		

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Type MS · Midpoints and points that split a line in a ratio

Answer 10 of 10 · Question 2 part c · 2 marks

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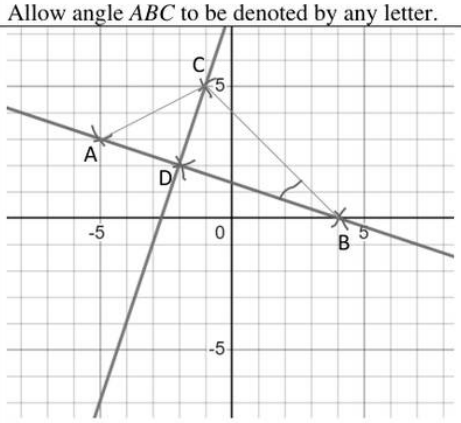
<table>
<tr><td rowspan="4">ALT 1</td><td>M1</td><td>Correct method to calculate one of the required lengths</td></tr>
<tr><td>M1</td><td>Correct method to calculate all of the required lengths</td></tr>
<tr><td>ddM1</td><td>Correct substitution into the cosine rule to give the expression shown, allow use of their lengths. Must obtain a correct expression for cosABC = , using their values.
Dependent on the previous 2 method marks.</td></tr>
<tr><td>A1</td><td>For $\left(\tan\angle ABC\right)=\frac{1}{2}$</td></tr>
<tr><td colspan="3">Because of the wording of the question, in not specifically asking for an exact value, we are going to allow a range for the value of $\tan\angle ABC$ as follows: $0.49989 \le \tan\angle ABC \le \frac{1}{2}$</td></tr>
<tr><td colspan="3">If you see the candidate using the ratio for a right angled triangle - but they are using the wrong pairing of lengths, (perhaps AB, BC or AC) then this is to be marked using the main scheme, and is unlikely to be awarded any marks unless they have correctly calculated BD or CD.

To be clear - If the candidate states correct values for AB, BC or AC but goes onto to use a tan ratio - this cannot be awarded any marks using the ALT mark scheme, because they are not using the correct cosine rule and with the tan ratio they should be using at least one of the correct sides, BD or CD.

If you see a clear attempt to use the cosine rule, please use this ALT marking scheme.

A special case is when the candidate has not made it clear which route they were intending to take - tan ratio (right angled triangle ratio) or cosine rule. If they have all lengths stated correctly, they could be awarded M1 M1 using either scheme.</td></tr>
<tr><td rowspan="2">Further ALTS</td><td></td><td>If you see a valid method, leading to a fully correct exact value for tanABC, please award 4 marks.

If the value for tanABC is incorrect or approximate and there is some partially correct working, PLEASE SEND TO REVIEW.

Allow angle ABC to be denoted by any letter.</td></tr>
<tr><td></td><td> 

For reference
$BD=\sqrt{40}(=2\sqrt{10})$ $AD=\sqrt{10}$ $AB=\sqrt{90}=3\sqrt{10}$ $BC=\sqrt{50}=5\sqrt{2}$
$\sin ABC=\frac{\sqrt{5}}{5}$ $\cos ABC=\frac{2\sqrt{5}}{5}$</td></tr>
</table>

Type MS · Midpoints and points that split a line in a ratio

Answer 10 of 10 · Question 2 part c · 2 marks

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Question number	Scheme	Marks
	a	M1 A1